

Homework 12

Problem 2

(a) $Y^f = \{(-z, q) : z \in \mathbb{R}_+^2, q \leq f(z) = Az_1^\alpha z_2^\beta\}$.

(b) We have shown (in homework 11) that Y^f satisfies CR to scale if and only if f is h.o.d. 1, i.e., $\alpha + \beta = 1$. You can also show that for a homogeneous production function (note that $f(z) = Az_1^\alpha z_2^\beta$ is h.o.d. $\alpha + \beta$), Y^f satisfies NI to scale iff $\alpha + \beta \leq 1$; Y^f satisfies ND to scale iff $\alpha + \beta \geq 1$. - You can also get these results by using the definition of NI and ND.

(c)

Case 1:

If $\alpha + \beta > 1$, then the production function $f(z)$ satisfies increasing return to scale (i.e., double inputs, the output will be more than double). And thus the average cost for the output will decrease as we produce more product (as the prices for inputs are given/unchanged).

We can write profit as $\Pi = (p - AC) * Q$, where p is the price of output, and AC is average cost (total cost divided by output), and Q is output. When inputs are great enough, AC will be smaller than p (i.e., $(p - AC)$ is positive), and therefore Π will go to infinity as our inputs go to infinity.

In this case, supply correspondence, profit function, conditional factor demand correspondence are not well defined (do not exist).

Case 2:

If $\alpha + \beta = 1$, then the production function $f(z)$ satisfies constant return to scale (i.e., double inputs, double output). In this case average cost is constant - we can get this by solving the cost minimization problem

$$C(Q) = \min_{\{z_1, z_2\}} w_1 z_1 + w_2 z_2 \quad (1)$$

$$\text{s.t. } Az_1^\alpha z_2^\beta = Q \quad (2)$$

we get (in (3)-(6), we use $\alpha + \beta = 1$)

$$z_1 = \frac{Q}{A} \left(\frac{\alpha w_2}{\beta w_1} \right)^\beta \quad (3)$$

$$z_2 = \frac{Q}{A} \left(\frac{\beta w_1}{\alpha w_2} \right)^\alpha \quad (4)$$

$$C(Q) = \frac{Q}{A} \left[\left(\frac{\alpha}{\beta} \right)^\beta + \left(\frac{\beta}{\alpha} \right)^\alpha \right] w_1^\alpha w_2^\beta. \quad (5)$$

Therefore, the average cost $AC(Q)$ is constant:

$$AC(Q) = \frac{1}{A} \left[\left(\frac{\alpha}{\beta} \right)^\beta + \left(\frac{\beta}{\alpha} \right)^\alpha \right] w_1^\alpha w_2^\beta \equiv AC. \quad (6)$$

So, under the condition $\alpha + \beta = 1$:

Subcase 1: if $p > AC$, the firm can produce infinite quantity and earns infinite profit. Again, supply correspondence, profit function, conditional factor demand correspondence are not well defined (do not exit).

Subcase 2: if $p < AC$, the firm will produce zero quantity and earns zero profit. In this scenario, supply is zero, profit is zero, conditional factor demand is also zero.

Subcase 3: if $p = AC$, the firm will produce any quantity Q such that $Q \geq 0$ and earns zero profit; supply is Q , profit is zero, conditional factor demand is given by (3) and (4).

Case 3:

If $\alpha + \beta < 1$, the profit maximization problem is

$$\max_{\{z_1, z_2\}} \pi = pAz_1^\alpha z_2^\beta - w_1 z_1 - w_2 z_2. \quad (7)$$

Taking FOCs (with respect to z_1 and z_2) of (7), and solve FOCs we get (8) and (9):

$$z_1 = \left(\frac{pA\alpha^{1-\beta}\beta^\beta}{w_1^{1-\beta}w_2^\beta} \right)^{\frac{1}{1-\alpha-\beta}} \quad (8)$$

$$z_2 = \left(\frac{pA\alpha^\alpha\beta^{1-\alpha}}{w_1^\alpha w_2^{1-\alpha}} \right)^{\frac{1}{1-\alpha-\beta}} \quad (9)$$

$$Q = Az_1^\alpha z_2^\beta = \left(\frac{p^{\alpha+\beta} A \alpha^\alpha \beta^\beta}{w_1^\alpha w_2^\beta} \right)^{\frac{1}{1-\alpha-\beta}} \quad (10)$$

$$\pi = pQ - w_1 z_1 - w_2 z_2 = (1 - \alpha - \beta) \left(\frac{pA\alpha^\alpha\beta^\beta}{w_1^\alpha w_2^\beta} \right)^{\frac{1}{1-\alpha-\beta}} \quad (11)$$

So in this case, supply correspondence is given by (10), profit function is given by (11) - note that profit is positive, conditional factor demand correspondence is given by (8) and (9).

In all three cases, we can always solve the cost minimization problem (CMP), and get the cost function. Note that cost function is well defined no matter $f(z)$ is constant return to scale, decreasing return to scale or increasing return to scale. The CMP is the same as (1) and (2) above. Solving the CMP, we get

$$z_1 = \left[\frac{Q}{A} \left(\frac{\alpha w_2}{\beta w_1} \right)^\beta \right]^{\frac{1}{\alpha+\beta}} \quad (12)$$

$$z_2 = \left[\frac{Q}{A} \left(\frac{\beta w_1}{\alpha w_2} \right)^\alpha \right]^{\frac{1}{\alpha+\beta}} \quad (13)$$

And hence we get the cost function

$$C(w_1, w_2, Q) = \left(\frac{Q}{A} \right)^{\frac{1}{\alpha+\beta}} \left[\left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha+\beta}} + \left(\frac{\beta}{\alpha} \right)^{\frac{\alpha}{\alpha+\beta}} \right] w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} \quad (14)$$

Note that when $\alpha + \beta = 1$, (14) reduces to (5).

(d) To verify Hotelling's Lemma, we have use (8)-(11) to verify that

$$\frac{\partial \pi}{\partial p} = Q \tag{15}$$

$$\frac{\partial \pi}{\partial w_1} = -z_1 \tag{16}$$

$$\frac{\partial \pi}{\partial w_2} = -z_2 \tag{17}$$